KLH UNIVERSITY
HYDERABAD



PROJECT PRESENTATION

Department of Computer Science and Engineering
KLH Bachupally Campus, Hyderabad

Team Number
T.N. - 10

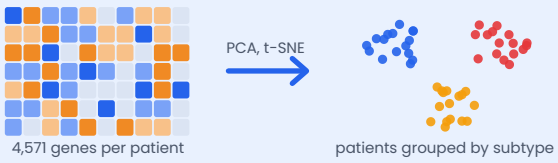
Team Members
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Guide Name
Mr. Shaik Asif

Project Title : **Visualizing Genomic Features for Disease Subtype Cluster Analysis**

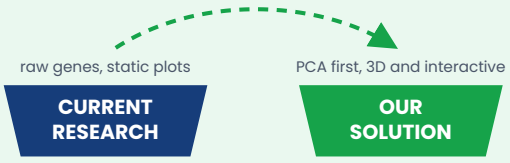
INTRODUCTION

- **Overview:** a machine learning pipeline that groups TCGA cancer patients into disease subtypes from gene activity (RNA-seq, 528 patients, 4,571 genes).
- **Problem statement:** thousands of noisy gene readings per patient make subtypes hard to see and slow to cluster.
- **Objectives:** reduce the data with PCA then t-SNE, group patients with K-Means and Hierarchical Clustering, and score separation and speed.
- **Scope and significance:** an interactive visual tool to explore subtypes and place new patients into a group.

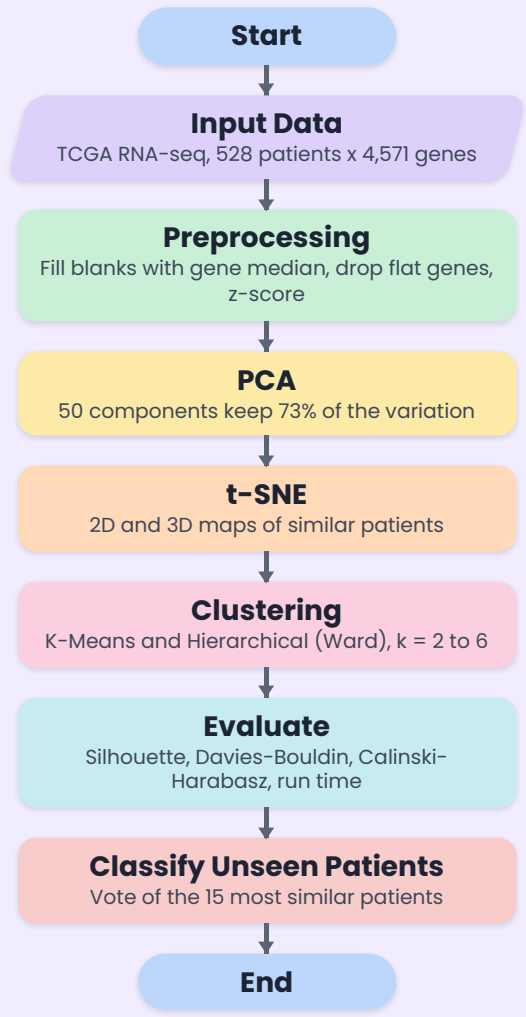


RESEARCH GAP

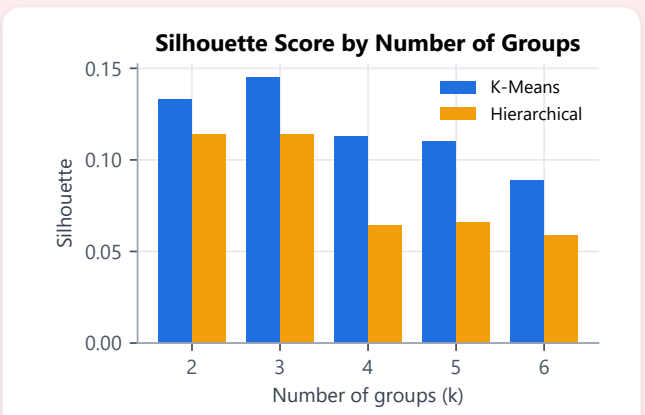
- **Limitations in existing approaches:** clustering raw gene data is slow and noisy, and results are usually shown as static plots.
- **Unexplored areas:** where a new, unseen patient fits, and how the method and number of groups change the result.
- **Need for improved solution:** a fast, visual way to compare methods and test the groups on held-out patients.
- **How our work addresses the gap:** PCA first pipeline (about 10x faster clustering), side by side scoring, a held-out test set and a 3D explorer.

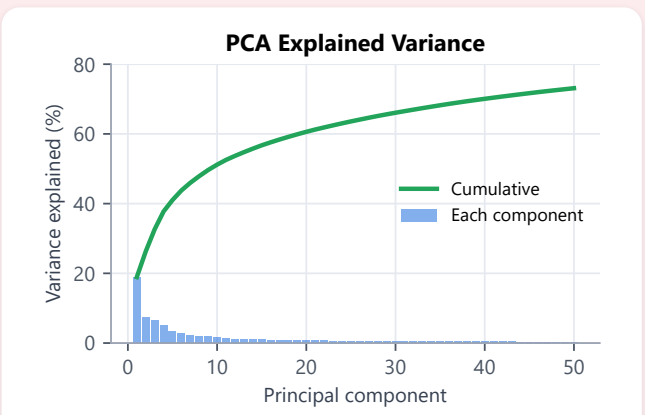


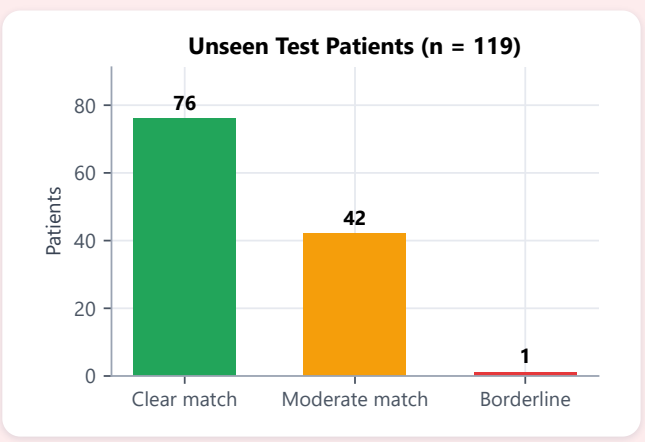
ALGORITHM

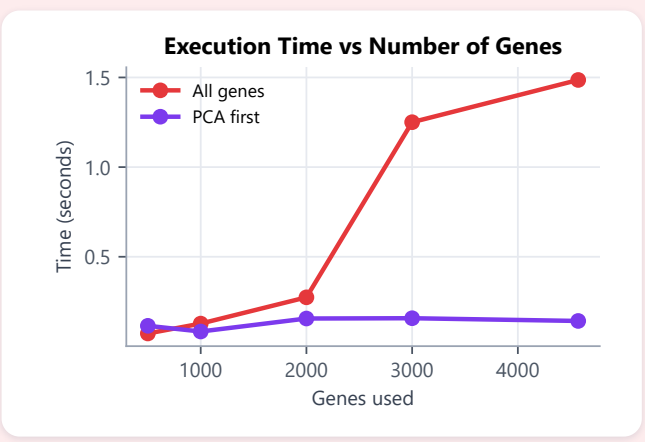


GRAPHS









Reading the results: K-Means separates the groups better than Hierarchical at every k, and the groups overlap (silhouette below 0.15). PCA first keeps clustering time flat while using all genes grows about 10x. 76 of 119 unseen patients land in a group with a clear match.